Edge AI Construction Site Safety Monitoring System

Advanced AI — Task 1 Technical Project Proposal

1.1 The Problem Statement (Integrated Version)

In heavy manufacturing, construction, and high-risk industrial sites, ensuring continuous workplace safety is an impossible task using manual human supervision alone. This reality exposes companies to massive operational, human, and financial risks:

1. **Human Supervision Bottlenecks & Monitoring Gaps:** Safety inspectors and security personnel cannot physically monitor sprawling, multi-tiered job sites or watch dozens of camera feeds continuously. Visual fatigue and staff limits mean manual inspections cover only a tiny fraction of total shift hours, creating extensive blind spots where violations go completely unseen.
2. **Behavioral Shortcuts Under Operational Pressure:** Under tight delivery deadlines, physical exhaustion, or harsh weather conditions, workers frequently take hazardous shortcuts to ease their tasks. This includes removing uncomfortable Personal Protective Equipment (such as hard hats, high-visibility vests, or safety harnesses) and bypassing basic safety protocols, exposing themselves to severe injuries or fatal accidents.
3. **Loss of Critical Personnel & Cascading Project Delays:** When an accident strikes, the disruption is immediate and severe. If a specialized technician or skilled operator is injured, their work grinds to a complete halt because specialized replacements are rarely available on short notice. A single injury can stall an entire critical path on the project timeline.
4. **Severe Financial & Legal Liabilities:** Workplace incidents carry devastating direct and indirect costs:
    - **Direct:** Heavy regulatory fines, workers' compensation payouts, and soaring future insurance premiums.
    - **Indirect:** Mandatory government investigations, site-wide Stop-Work Orders, liquidated damages (penalties) for delayed project delivery, and disqualification from bidding on future enterprise tenders.
5. **Information Asymmetry (Reactive vs. Proactive):** Currently, executive management and safety directors operate with zero real-time visibility; they only learn about safety failures *after* a catastrophic incident or an official audit penalty occurs. The industry fundamentally lacks an automated, real-time auditing system capable of detecting visual hazards early and intervening before a shortcut turns into a project-halting disaster.

1.2 Importance of Solving the Problem

Solving this problem is critical to transforming high-risk job sites from hazardous, high-liability zones into predictable, high-efficiency operations:

1. **Workforce Protection & Project Momentum:** Worker safety is an absolute operational baseline. In specialized construction and industrial projects, human capital is not instantly replaceable. A single critical injury or fatality does not just destroy a life—it halts site momentum, demoralizes the entire workforce, and triggers mandatory governmental stop-work investigations that freeze project milestones for weeks.
2. **Financial Continuity & Liability Defense:** Workplace accidents inflict severe direct and indirect financial damage. Beyond immediate medical compensation to victims and their families, companies face escalating insurance premiums and contractual liquidated damages (financial penalties for project delivery delays). Preventing incidents proactively protects the bottom line and preserves the contractor's safety rating required to win enterprise-level tenders.
3. **Real-Time Prevention vs. Passive Recording:** Traditional security teams and static CCTV systems suffer from human latency and visual fatigue; they merely document a disaster after it occurs. Automating detection bridges the gap between passive recording and active prevention. By delivering instantaneous automated alerts the moment a violation happens (such as entering a hazardous zone or shedding PPE), the system empowers site supervisors to intervene *before* an unsafe behavior turns into an irreversible casualty.

1.3 Affected Entities & Stakeholders

Globally, this problem is experienced across the entire construction, heavy manufacturing, and industrial sectors. The impact directly falls on:

1. **Construction & Industrial Companies Worldwide:** From local contractors to multinational infrastructure developers, companies constantly face operational disruptions, project-halting investigations, regulatory fines, and escalating compensation costs due to unmonitored safety violations.
2. **On-Site Field Workers:** Frontline laborers, technicians, and equipment operators who bear the direct physical cost of safety blind spots, suffering preventable injuries, permanent disabilities, or fatal accidents.
3. **Site Safety Officers & Project Supervisors:** Field supervisors who are held legally and operationally accountable for site compliance, but lack the physical capacity to maintain continuous oversight across active, multi-zone job sites.

1.4 Current Solutions

Currently, organizations tackle site safety monitoring through three primary approaches:

1. **On-Site Safety Supervision (HSE Teams):** Hiring dedicated safety managers and site supervisors who conduct regular manual walkthroughs, inspect work zones, and enforce PPE compliance on foot.

2. **CCTV Control Rooms & Video Monitoring:** Installing closed-circuit cameras (CCTV) linked to on-site security control rooms, where personnel observe multiple video feeds to spot safety breaches and hazardous behavior.
3. **Emerging AI & Computer Vision Solutions:** A growing number of companies have begun adopting automated AI models to assist human teams—monitoring live video feeds to detect missing PPE and extend visual coverage across active work areas. *(The operational limitations of these current AI offerings and our distinct competitive edge will be detailed in the following sections).*

1.5 Limitations of Current Solutions

While existing safety methods aim to enforce compliance, they suffer from critical human, operational, and technical constraints:

1. **Human Cognitive Fatigue & The "Cat-and-Mouse" Dynamic:** Manual inspection relies on human vigilance, which rapidly deteriorates due to visual fatigue, distraction, and sprawling site geography. Furthermore, it creates a "cat-and-mouse" enforcement culture—workers wear their PPE only while an inspector is physically present and immediately shed it once the inspector leaves, leaving long intervals of unmonitored risk.
2. **Subjective Enforcement & Conflict of Interest:** Human supervision is vulnerable to personal discretion, leniency, or external operational pressure. When faced with aggressive delivery deadlines or cost constraints, on-site personnel may intentionally overlook safety shortcuts to prioritize speed over protocol compliance, resulting in compromised site standards.
3. **The "Forensic" Lag of Traditional CCTV:** Standard video monitoring is passive and forensic; security staff rarely spot subtle non-compliance across dozens of simultaneous screens. Consequently, footage is almost exclusively reviewed *after* an accident occurs to investigate liabilities, completely failing as a proactive, life-saving mechanism.
4. **Barriers in Existing Commercial AI Tools:** Early-stage automated safety systems in the market remain largely inaccessible to most contractors due to exorbitant enterprise licensing fees, complex proprietary hardware dependencies, and a high frequency of false-positive alarms caused by harsh industrial environmental conditions (such as dust, dynamic shadows, and heavy equipment obstructions).

2.1 Proposed Solution & Platform Capabilities

The system operates as an autonomous, computer-vision-based safety intelligence platform designed for dynamic construction environments. Its primary function is to transform passive CCTV infrastructure into an active, real-time risk prevention mechanism across three distinct operational layers:

- **Automated PPE Compliance Verification:** Continuously audits personnel for mandatory Personal Protective Equipment (PPE)—including hard hats, high-visibility vests, and fall protection harnesses. The system flags missing gear immediately upon entry or during active work in designated zones.

- **Behavioral Anomaly & Unsafe Action Detection:** Analyzes human posture and spatial dynamics to identify active protocol breaches. This includes detecting workers in prohibited equipment swing radiuses, tracking unauthorized entry into high-risk geofenced zones, and identifying unsafe maneuvers (e.g., improper ladder use, leaning over unsecured edges).
- **Environmental & Situational Hazard Recognition (Predictive Safety):** Monitors the site topology to detect evolving environmental risks independent of direct worker actions. It identifies static hazards (e.g., uncovered openings, unbarricaded edges, unstable material stacking) and maps active work activities against ambient hazards to proactively alert teams before an operation leads to an incident.
- **Real-Time Notification Pipeline:** Generates targeted, low-latency alerts to site safety officers and supervisors via a centralized dashboard and mobile/radio triggers, attaching visual proof (bounding boxes and timestamps) to prevent disputed infractions.

2.2 Computer Vision Intelligence & Models

Computer Vision replaces reactive, intermittent manual supervision with deterministic, low-latency visual intelligence. Instead of relying on human inspectors who are constrained by fatigue, blind spots, and site scale, CV continuously parses video streams at the frame level using specialized models:

- **Real-Time Object Detection (PPE Auditing):** Convolutional networks and transformer-based vision detectors (e.g., YOLO/RT-DETR variants) perform high-frequency bounding-box localization and classification. The models detect required PPE (hard hats, high-vis vests, harnesses) and flag absent gear under complex environmental noise (dust, shadows, varied worker distances).
- **Pose Estimation & Kinematic Analysis (Behavioral Safety):** Keypoint detection algorithms map worker skeletal topology to infer unsafe body postures, balance loss, or hazardous maneuvers (e.g., reaching over unprotected voids, improper lifting techniques) without requiring wearable physical sensors.
- **Multi-Object Tracking (MOT) & Dynamic Geofencing:** Spatial tracking algorithms calculate relative proximities and trajectories between workers and heavy machinery (e.g., excavators, mobile cranes). By defining programmatic safety boundaries, CV triggers spatial intrusion alerts whenever a worker breaches an active danger zone.
- **Local Edge Inference (Zero-Latency Guarantee):** By running quantized models directly on site-deployed edge gateways, inference occurs locally within tens of milliseconds. This eliminates cloud bandwidth dependencies and guarantees actionable, split-second intervention before an incident materializes.

2.3 System Inputs

To operationalize the safety monitoring platform, the site operator/client provides the following inputs:

- **Live Video Streams (RTSP / ONVIF):** Continuous, low-latency video feeds from existing or newly deployed IP cameras, mobile solar-powered CCTV towers, and crane-mounted lenses covering critical work zones.

- **Spatial Configuration & Geofencing Boundaries:** Site layout blueprints or visual zoning markup on the management interface to delineate active hazardous zones, equipment swing radii, overhead lift corridors, and exclusion areas.
- **Safety Protocol Matrix:** Site-specific compliance rules specifying mandatory PPE requirements mapped by zone (e.g., mandatory safety harnesses in elevated zones vs. standard boots/vests in ground staging areas) and operational work schedules.
- **Alert Escalation Directory:** Designated routing tables for site personnel, including phone numbers, radio channels, and device IDs for field safety officers and supervisors to ensure instant incident dispatch.

2.4 System Outputs

The platform delivers deterministic, actionable outputs categorized into real-time operational alerts and strategic compliance intelligence:

- **Low-Latency Multi-Channel Incident Dispatch:** Instantaneous push alerts dispatched to designated field safety personnel via SMS, mobile application, or integrated radio channels upon detection of a critical breach. Alerts include timestamps, specific site zones, severity levels, and captured frame snapshots with annotated bounding boxes.
- **Augmented Real-Time Video Feeds:** Live visual streams overlaid with analytical visual telemetry—highlighting missing PPE, dynamic danger geofence breaches, and color-coded proximity vectors between heavy machinery and ground personnel.
- **Centralized Safety Dashboard & Risk Heatmaps:** A unified web/mobile interface providing safety managers with continuous situational awareness, including real-time PPE compliance rates, recurring near-miss trend analysis, and spatial heatmaps that pinpoint chronic hazard zones across the job site.
- **Immutable Compliance & Audit Logging:** Automated, tamper-proof event logs detailing every detected anomaly, resolution status, and supervisor acknowledgement. These verifiable audit trails serve as legal documentation to counter disputed claims and support insurance premium discount negotiations.

2.5 Differentiators & Workflow

Traditional construction safety relies either on manual Health, Safety, and Environment (HSE) site inspections or passive, forensic CCTV surveillance. The proposed Edge-AI system resolves the structural failures inherent in these methods through five distinct differentiators:

- **Elimination of the "Observer Effect" via 24/7 Site-Wide Persistence:**
  Manual audits suffer from intermittent sampling; workers routinely wear PPE only when safety officers are physically present and discard it once they move away. The AI system provides continuous, ubiquitous monitoring across all active zones, creating a sustained culture of compliance with zero spatial or temporal blind spots.
- **Deterministic, Sub-Second Intervention (Zero Latency):**
  Human oversight and traditional CCTV operate retroactively—reviewing footage after an incident occurs to assign liability. By processing video feeds at the edge (

inference time), the system intercepts precursors to accidents and dispatches immediate site alerts before an unsafe condition escalates into a fatality.

- **Immunity to Cognitive Fatigue, Distraction, and Bias:**
  Human attention spans degrade significantly under severe site conditions (heat, glare, repetitive visual stimuli, multi-tasking). The computer vision pipeline executes inference with uniform precision around the clock, unaffected by fatigue, interpersonal bias, or the operational pressure to overlook minor infractions to hit deadlines.
- **Objective and Immutable Compliance Verification:**
  Human incident logs are inherently subjective and frequently disputed in regulatory audits or insurance claims. The platform replaces anecdotal logs with cryptographically verified, timestamped telemetry and annotated frame captures, eliminating human error in reporting.
- **Exponential Scalability as a Workforce Multiplier:**
  Instead of requiring contractors to hire dozens of safety officers to monitor expanding multi-acre construction sites, the system scales across hundreds of camera feeds simultaneously. This transforms safety officers from manual patrol wardens into high-leverage responders who act exclusively on verified, high-risk hazards.

System Workflow

- **Input Stage:**
  The pipeline captures synchronized live video feeds (15–30 FPS) from edge-mounted IP cameras. Concurrently, it ingests site boundary geometry, active work shift parameters, and location-specific PPE rules configured by site management.
- **Computer Vision Model (Edge Inference):**
  Raw video frames undergo hardware-accelerated preprocessing directly on local industrial edge gateways. Quantized deep learning vision models execute parallel inference: detecting PPE gear, mapping human skeletal joints to recognize dangerous postures, and tracking workers across moving machinery paths.
- **Processing & Decision Logic:**
  Inference metadata passes into a rule-based decision engine. To eliminate spurious false alarms (e.g., brief sensor occlusions or momentary head turns), the engine requires violations to persist across multiple consecutive frames (-frame temporal check). It calculates Euclidean spatial proximities and matches detected infractions against pre-set severity tiers (e.g., missing vest = Tier 2; entering active crane radius = Tier 1 Emergency).
- **Output Stage:**
  Upon confirming an actionable violation, the system executes low-latency dispatch: sending targeted push alerts with captured visual crops to field supervisors, updating live web dashboards with site heatmaps, and appending the event to an immutable audit ledger for compliance records and insurance verification.

3. Business Overview

Target Customers

Our main target customers are businesses with active, high-risk work environments:

- **1. Construction Companies**
    - They manage large work sites with heavy machinery and hundreds of workers.
    - They need an easy way to make sure everyone wears safety gear (helmets, vests) and stays away from dangerous areas.
- **2. Factories & Industrial Plants**
    - They operate indoor facilities with moving equipment (like forklifts) and hazardous zones.
    - They need 24/7 monitoring to prevent accidents and follow basic safety rules.

Value Proposition

Our solution provides clear, practical value by transforming how safety is managed on-site:

- **Improves Safety:** Protects workers in real time by catching missing safety gear and dangerous actions immediately, preventing injuries and fatal accidents before they occur.
- **Automates a Manual Process:** Replaces slow, physical site walks with automated camera monitoring that watches every corner of the site continuously.
- **Reduces Human Error:** Unlike human supervisors who can get tired, distracted, or overlook things in the heat, the computer vision system never loses focus and applies safety rules equally to everyone.
- **Improves Quality:** Raises the overall quality of workplace operations by enforcing high safety standards consistently 24/7, creating a cleaner, safer, and more organized environment.
- **Saves Time & Reduces Cost:** Saves inspectors hours of repetitive patrols and saves the company money by preventing costly work shutdowns, legal fines, and high insurance premiums.

Business Model

We use a **B2B (Hardware + Software)** hybrid model:

- **Hardware (One-time Setup Fee):** Selling or leasing the local **Edge Processing Units** (the hardware that runs the AI models on-site) and any required cameras.
- **Software Subscription (Recurring B2B SaaS):** A monthly or annual subscription fee per site (or per active camera) to access the AI detection software, the monitoring dashboard, instant alert system, and ongoing technical support.

Competitors

1. Intenseye

- **What do they offer?**
  - An enterprise AI workplace safety platform designed to transform existing CCTV cameras into 24/7 safety monitors across manufacturing and industrial facilities.
  - Provides safety managers with automated violation tracking, near-miss logging, and compliance dashboards.
- **How do they use Computer Vision?**
  - **PPE Auditing:** Detects missing hard hats, vests, safety glasses, and gloves.
  - **Area & Zone Monitoring:** Flags workers crossing into restricted areas or walking into danger zones.
  - **Vehicle-Pedestrian Proximity:** Identifies unsafe distances between forklifts/machinery and pedestrians.
  - **Ergonomics & Behavior:** Analyzes human posture to identify risky lifting and body strain.
- **Strengths:**
  - Offers 50+ pre-trained EHS (Environmental Health & Safety) use cases out of the box.
  - Built-in privacy features that automatically blur faces and skin to protect worker identity.
  - Integrates easily with existing IP cameras without requiring new cameras.
- **Limitations:**
  - **Heavy Cloud Reliance:** Relies mainly on cloud processing, which requires stable, high-speed internet—often unavailable on raw construction sites.
  - **Latency:** Cloud round-trips can cause slight alert delays, making split-second hazard warnings difficult.
  - **High Enterprise Cost:** Targets large Fortune 500 enterprises with expensive enterprise pricing plans.

2. Protex AI

- **What do they offer?**
  - A video analytics platform focused primarily on capturing proactive safety data, near-miss events, and risky behaviors in logistics, ports, and warehouse environments.
- **How do they use Computer Vision?**
  - **Object Tracking:** Tracks interactions between mobile warehouse equipment (like forklifts) and walking workers.
  - **Geofencing & Path Tracking:** Checks whether workers use designated pedestrian walkways or cross dangerous machinery paths.
  - **Near-Miss Identification:** Uses spatial calculations to detect close calls before an actual injury happens.
- **Strengths:**
  - Very strong focus on proactive "near-miss" reporting rather than just catching missing gear.

    - Privacy-by-design architecture that anonymizes all captured video feeds.
    - Intuitive and clean reporting dashboard for safety managers.
- **Limitations:**
    - **Structured/Indoor Focus:** Primarily optimized for organized indoor settings (warehouses and factories) rather than messy, rapidly changing outdoor construction sites.
    - **Limited Offline Capability:** Still depends heavily on cloud connectivity for centralized model management and processing.

3. What Could Your Solution Do Differently?

- **Edge-First Processing (Works Offline with Zero Latency):**
    - Unlike competitors who stream heavy video to the cloud, your system processes frames locally on an **Edge Box** on-site. It alerts workers in milliseconds and keeps running even when the construction site has poor or no internet.
- **Built Specifically for Dynamic Outdoor Sites:**
    - Tailored for rough construction environments (dust, sun glare, moving cranes, scaffolding, and changing site layouts) rather than clean indoor warehouses.
- **Direct On-Site Interventions:**
    - Integrates directly with site alarms, sirens, and radios to stop workers right at the moment of danger, not just sending an email report hours later.
- **Cost-Effective B2B Model:**
    - Provides an accessible hardware + software package designed for local and regional contractors who cannot afford high-end enterprise platforms.

4. Technical Specifications & Computer Vision Pipeline

Computer Vision Tasks

Our project is a multi-task vision pipeline that combines the following Computer Vision problems:

- **1. Object Detection**
    - **Role:** Locating and identifying multiple objects in each video frame with bounding boxes.
    - **Use Case:** Detecting workers, machinery (cranes, excavators), and mandatory PPE (hard hats, high-vis vests, safety harnesses).
- **2. Object Tracking**
    - **Role:** Assigning a persistent ID to detected workers and equipment across consecutive frames over time.
    - **Use Case:** Tracking worker movement to verify if someone enters a dynamic danger zone (like the swing radius of a crane) and calculating distances between workers and moving machinery.
- **3. Pose Estimation**

    - **Role:** Mapping human body joints and skeletal keypoints.
    - **Use Case:** Detecting hazardous postures, slips, trips, leaning over open edges, and improper climbing on ladders without relying on wearable sensors.
- **4. Image Segmentation**
    - **Role:** Pixel-level classification of areas within the camera's view.
    - **Use Case:** Defining precise boundaries for dynamic hazard zones (like floor openings, excavation pits, and designated safe walkways) rather than simple rectangular boxes.

Data Requirements

What data would you need?

1. **Data:** Images of workers wearing and NOT wearing safety gear.
2. **Hazardous Situations:** Images of unsafe conditions (uncovered holes, unprotected edges).
3. **Behavioral Videos:** Short video clips showing risky movements and equipment interactions.

What would the images/videos contain?

- **1. Workers with & without PPE:** Clear footage of workers on-site—some wearing full safety gear (hard hats, vests, harnesses) and others missing specific items, captured from different camera angles.
- **2. Unsafe Site Conditions:** Visual examples of hazardous areas, such as open floor edges without guardrails, uncovered pits, and unstable scaffolding.
- **3. Dangerous Interactions & Near-Misses:** Short video sequences showing risky behavior before accidents happen (e.g., workers walking inside a crane's blind spot or stepping too close to heavy moving machinery).

How Would You Collect the Data?

- **1. Existing Public Datasets:** Use verified open-source construction datasets (e.g., from Roboflow or Kaggle) to quickly get thousands of labeled images for helmets, vests, and workers.
- **2. Real Site Footage (On-site Collection):** Collect video recordings from active CCTV cameras and mobile towers across partner construction sites to capture real-world conditions.

Would you use an existing dataset?

Probably yes

Would you need to create your own dataset?

Also yes I would need to

Approximately how much data might be required?

Probably 10K photos and 100 videos as we r using pretrained models as yolo so we won't need too much data

Relevant dataset:

Construction Site Safety Image Dataset Roboflow on kaggle

Possible Computer Vision Models

1. Object Detection Models (Core)

- **YOLO**
  - **Why:** Single-stage detector with optimal balance between inference speed and mAP (mean Average Precision). It runs efficiently on Edge devices (like NVIDIA Jetson) using TensorRT optimization.
- **Faster R-CNN**
  - **Why excluded:** Two-stage detector. Very accurate, but far too slow (low FPS) for real-time safety alerts on edge devices.
- **MobileNet-SSD**
  - **Why considered:** Ultra-lightweight and extremely fast, but struggles with detecting small objects from a distance (like a helmet viewed from a high CCTV camera).

2. Multi-Object Tracking Models (Temporal Continuity)

- **ByteTrack**
  - **Why:** Fast, lightweight, and associates low-score detection boxes to preserve track IDs during occlusions (e.g., worker moving behind scaffolding or equipment).

3. Pose Estimation & Spatial Analysis

- **YOLOv8-Pose / YOLO11-Pose:**
  - Maps 17 body keypoints to detect falls, awkward postures, or slipping without needing a separate model architecture.

Possible Architecture

1. Single-Stage Unified Architecture (Direct End-to-End)

- **Structure:** Single Deep Neural Network (Backbone Feature Neck Detection Heads) running in a single forward pass (e.g., YOLOv8 / YOLO11).
- **Workflow:** Takes the full camera frame as input and simultaneously predicts bounding boxes and classes for all objects: Worker, Hardhat, No-Hardhat, Vest, No-Vest, and Machinery.
- **Pros:**

    - Lowest latency (fastest FPS).
    - Fixed processing time regardless of how many workers are present in the frame.
    - Best fit for constrained **Edge AI** hardware.
- **Cons:**
    - Can struggle with small objects (e.g., detecting a hardhat when a worker is far away in a high-mounted CCTV view).

Traditional Computer Vision Techniques

Geofencing & Spatial Mathematics

- **Ray-Casting (Point-in-Polygon):** Determines whether worker bounding-box coordinates fall inside irregular hazard zones (excavation pits, crane swing radii).
- **IoU (Intersection over Union):** Evaluates bounding-box overlaps between workers and heavy machinery to detect proximity violations.

Deep Learning Techniques & Architectures

Single-Stage CNN Detectors (YOLO Family — CSPDarknet Backbone + PANet Neck)

- **Architecture:** Uses Cross-Stage Partial networks (CSPDarknet) to extract multi-scale spatial features, combined with Path Aggregation Networks (PANet) to preserve small-object details (e.g., helmets viewed from high-angle CCTV cameras).
- **Role:** Real-time object detection (workers, machinery, safety gear) in a single forward pass

Tools & Technologies

- Python
- Opencv
- Numpy
- Pytorch
- Yolo
- Cnn
- Docker

5. Feasibility Analysis

Is this project technically possible?

Yes it is

What are the biggest technical challenges?

To make it work locally without internet or hosting it on cloud

What type of hardware might be required?

Central system that could hold the CV model so we could perform the offline part and the cameras connected to the central system and an alert system

What data problems could you face?

Incidents like near-collisions between workers and heavy machinery are rare occurrences, making it difficult to collect balanced positive training samples for the training data

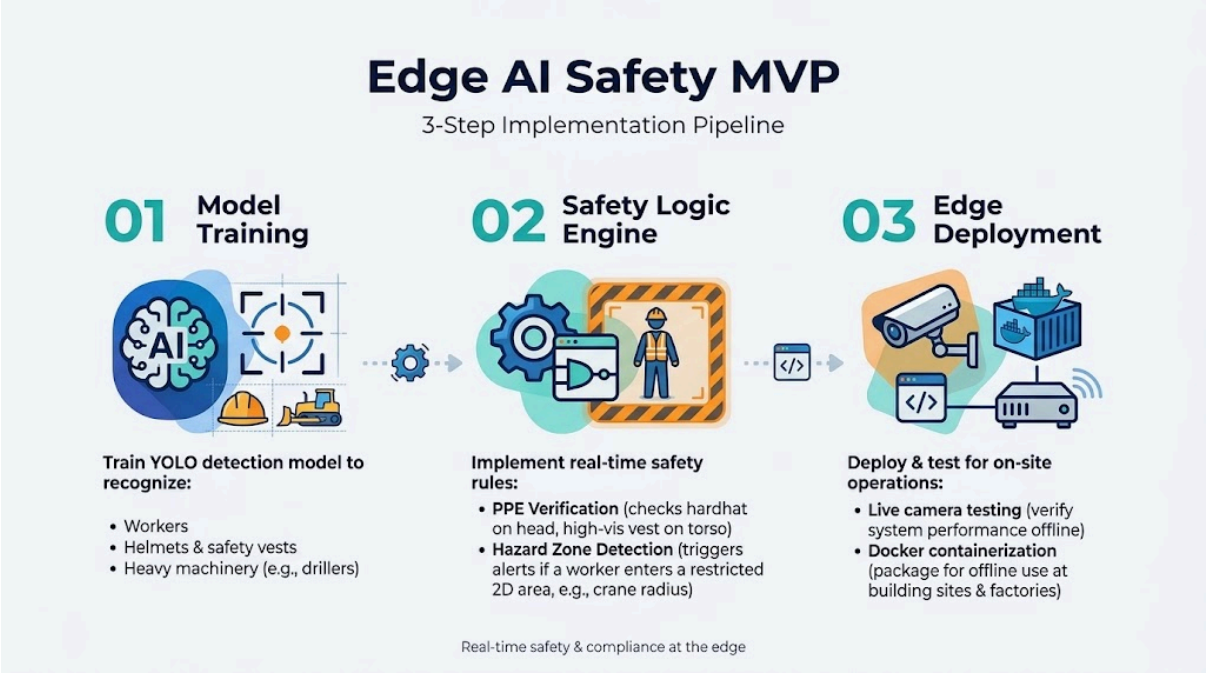
What could make the model inaccurate?

Sunlight, dust , rapid movement of workers that could make interruption in the frame rate, workers not visible in construction sites or wearing clothes the same color as the safety kit

What limitations would your first version have?

6. 1-**Limitation:** V1 will operate each camera feed as an independent, isolated processing pipeline.
    - **Impact:** The system cannot perform cross-camera worker re-identification across multiple site zones; a worker moving from Zone A to Zone B will be logged as two separate, uncorrelated subjects.
7. 2-V1 will be trained and evaluated primarily on daytime footage under moderate lighting conditions so nightshifts, raining time or foggy times would make a great problem for the system
8. 3-Reliable detection will be constrained to workers within roughly **15 to 25 meters** of the camera. Workers positioned farther away in wide-angle views will appear too small for confident helmet/vest identification.

6. MVP & Architecture



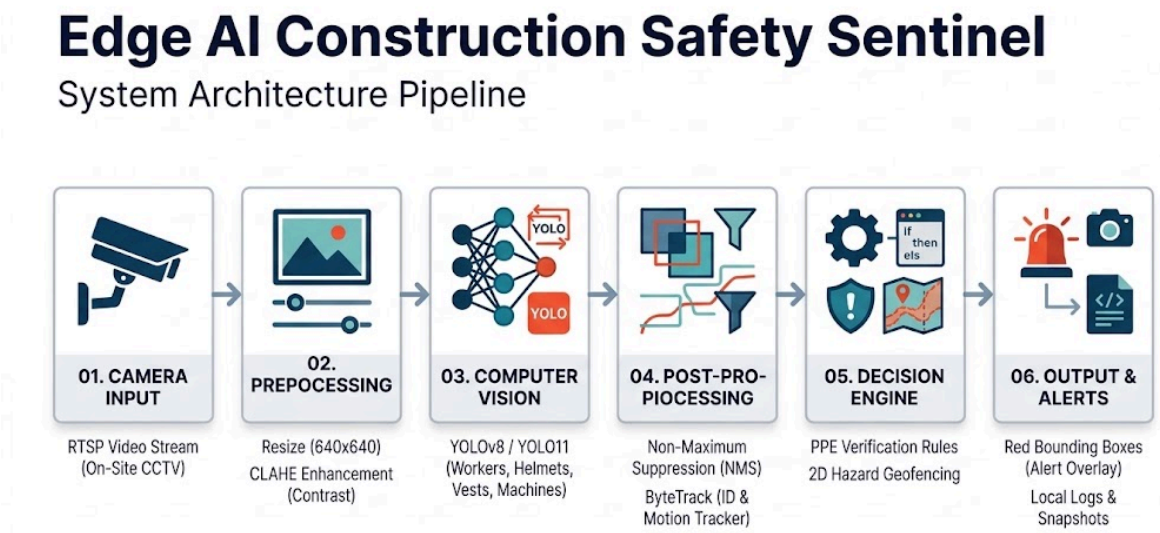
First we train yolo model to recognize(workers, helmet, safety vest and heavy machines as drillers).

Second we make the logic which is:

- **PPE Verification:** Checks if a detected worker has a hardhat on their head region and a high-visibility vest on their torso.
- **Hazard Zone Detection:** Trigger an alert if a worker steps into a restricted area (e.g., crane radius or open excavation).

And lastly we check that the system is working fine by connecting it to a camera and trying it then deploying it to a docker file to use it offline in building sites and factories.

Final Project Architecture



1. Camera / Image

- **RTSP IP Camera:** Ingests live 1080p video streams directly from the on-site CCTV camera.

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2. Preprocessing

- **OpenCV Resizing & Enhancement:** Resizes incoming frames to 640x640 and applies contrast adjustments (CLAHE) to handle site lighting conditions.

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3. Computer Vision Model

- **YOLO (v8 / 11):** Performs real-time object detection in a single forward pass to detect workers, helmets, safety vests, and machinery.

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4. Post-processing & Tracking

- **NMS & ByteTrack:** Removes duplicate overlapping boxes using Non-Maximum Suppression (NMS) and assigns persistent tracking IDs to workers across consecutive frames.

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5. Decision / Prediction (Safety Engine)

- **Safety Logic Verification:**

- **PPE Compliance:** Verifies that a hardhat is present on the worker's head region and a safety vest covers the torso.
- **Hazard Geofencing:** Determines whether worker bounding-box coordinates intersect with predefined restricted danger polygons.

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6. Application / Output

Local Alerts & Offline Logging: Overlays red alert bounding boxes on the video feed and saves timestamped violation snapshots locally for supervisor review.

Research Sources

2 technical sources

1. https://onlinelibrary.wiley.com/doi/10.1155/2020/9703560
2. https://docs.opencv.org/5.0/

Data source

1. https://www.kaggle.com/datasets/snehilsanyal/construction-site-safety-image-dataset-roboflow

2Competitors

1. https://www.protex.ai/
2. https://www.intenseye.com/